Anirudh Sarma Kolluru

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Education

Purdue University ~ West Lafayette, IN
B.S. Mechanical Engineering ~ 3.51 GPA
Minor Business Economics

Experience

Gas Engineering Intern | Pacific Gas and Electric Company — Jun 2026 – Aug 2026
- Developed a program that automatically imports files for pre-inspection analyzing and delivering most important information automatically filtering to anomalies within equipment handling over **20,000+ equipment** reducing time from **2 weeks to 3 minutes**
- Built a PowerApps Form that streamlines the Notice of Violation process allowing for clean user input minimizing all human error

Undergraduate Research Fellow | Manufacturing for Extremes Lab — Aug 2025 – *Present*
- Designed grips to perform creep tests on materials below **ASTM** standards using **Siemens NX** validating designs with **FEA**
- Produced technical drawings and applied **GD&T** to ensure successful part manufacturing
- Created a Digital Acquisition program that would be able to control and adjust a creep controller in real time using Python and MatPlotLib

Machine Shop Machinist | ME Project Machine Shop — Aug 2025 – Jan 2026
- Worked at the Machine Shop assisting students in operating machining equipment
- Assisted **100+** students in using the Hydraulic Press, Mill, and Lathe ensuring adherence to safety protocols
- Applied engineering drawings and tolerances to ensure manufacturing accuracy

Product Development Intern | Karlyan — May 2025 – Aug 2025
- Designed an end-to-end mechanical heliostat with **Solidworks** bringing sunlight into areas without needing electricity
- Applied **DFM** by changing material specification and reducing parts while still maintaining **3.6** Factor of Safety
- Performed cost-benefit analysis to adapt product for rural areas while also decreasing overall energy consumption by **37%** in urban areas

Aerodynamics Sub Team Member | Purdue Formula SAE — Aug 2024 – Jan 2026
- Utilized Computational Fluid Dynamics **(CFD)** to design and optimize a high-performance rear wing with a CL of **-2.7**
- Conducted tests through **Ansys** to determine downforce and reduce radiator drag on the **undertray (UT)**
- Collaborated with other members to design and mold undertray out of carbon fiber

Engineering Intern | ECSite — Jun 2024 – Aug 2024
- Designed a scalable automated UIPath program to manage **1700+** equipment processes
- Optimized overall scraping program runtime by **20%** from 5 to 4 days
- Programmed a **MATLAB** script to visualize antenna **(DAS)** signal propagation to find weak spots in given systems

Projects

Robotic Arm | Robotic/Controls Design — Jun 2026 – *Present*
- Designing a 3d printable robotic arm using Siemens NX fit with servo motors to enable 3 degrees of freedom
- Created an OpenCV program that can capture arm movements sending commands to the arm to mimic
- Developed a LQR (Linear Quadratic Regulator) Controls algorithm to ensure swift smooth motion of arm

Solar Farm Optimizer | Machine Learning — Jan 2026 – May 2026
- Programmed a Machine Learning Framework that analyzes any given image delivering the optimal solar farm placement area
- Uses a Random Forest Model with Gaussian Noise to incorporate pixel variety to allow for a range in pixel values
- Scores map through a 15 dimensional feature space to determine the land type

RC Airplane | Aerospace Engineering — Jun 2025 – Aug 2025
- Created a new fuselage and airfoil design performing CFD analysis to improve aerodynamics
- Executed controls analyses to ensure stable flight patterns

Audio Equalizer | Electrical Engineering — Apr 2025 – May 2025
- Used multiple Op-Amp Comparators and ICs **(LMC 324&555)** to create a working audio equalizer and working siren
- Ran signal processing analyses to create functioning band pass circuits with error below **~5%** using **LTSpice**

Skills

Programming
- Java, C++, C, Python, MATLAB
- Machine Learning
- HTML & CSS

Engineering
- NX, Fusion 360, SolidWorks
- CFD & FEA Analysis
- EE Design/Breadboarding
- Thermal/Fluids/Controls Design
- Manual Machining
- FDM 3D Printing